

Project Report

Multi-Class Legal Citation Classifier

- A Machine Learning Approach to Precedent Analysis

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1. Introduction

This project addresses a fundamental challenge in legal practice: automatically discerning the **nature of judicial reliance** on prior cases (precedents). While Information Retrieval systems can locate a cited case, they often fail to classify *how* that case was used. This classification is vital under the doctrine of **Stare Decisis** (precedent).

The project's objective was to build a machine learning model capable of performing **Multi-Class Classification** on raw legal text. The goal is to classify document segments into one of four categories: *cited*, *followed*, *referred to*, or *other* (non-citation outcome). This model is designed to serve as the core AI component of a modern Legal Research Assistant.

The project adhered strictly to a one-week timeline, focusing on demonstrating the full pipeline of an Artificial Intelligence and Data Science workflow, from raw data ingestion to final model evaluation.

2. Literature Survey

The application of Artificial Intelligence to legal text, often termed **LegalTech** or **Legal NLP**, is a rapidly evolving field. Legal texts present unique challenges due to their highly structured, verbose, and context-dependent nature.

Traditional NLP Baselines: Many foundational Legal NLP projects rely on established techniques like **Term Frequency-Inverse Document Frequency (TF-IDF)** combined with simple yet effective classifiers such as **Logistic Regression** and **Support Vector Machines (SVM)**. These methods, which capture the importance of specific vocabulary, have proven to be robust starting points for text categorization tasks.

Citation Classification: Academic research confirms that distinguishing between subtle citation categories (like cited vs. followed) requires robust pre-processing. The difference often hinges on modal language (e.g., "the court **must** follow" vs. "the court **may** refer to") rather than just topical keywords. This necessitates aggressive text normalization techniques like **Lemmatization** to ensure all inflections of a critical verb (like 'follow') are consolidated.

The Data Challenge (Imbalance): A pervasive issue in legal datasets is **class imbalance**. Not all judicial actions occur with the same frequency. A small

fraction of cases will be definitively 'followed,' while the vast majority will be merely 'cited' or 'distinguished.' This imbalance directly impacts machine learning performance, leading to a bias toward the majority class, a critical factor analyzed in this report.

3. Methodology

The project followed a four-phase methodology

3.1. Data Acquisition and Preparation (Phase 1)

The raw dataset contained 25,152 documents.

- **Feature Selection:** The column **case_text** was selected as the raw text input.
- **Target Creation:** The project focused on the **case_outcome** column, which contained the citation type as a string. A data transformation step mapped the specific citation words (cited, followed, referred to) to distinct classes, while grouping all other outcomes (e.g., vacated, affirmed) into a single **other** class.
- **Data Imbalance:** The target variable distribution confirmed a severe imbalance, which influenced all subsequent phases (Table 1).

Table 1: Multi-Class Target Distribution

CITATION_TYPE	Count
cited	12,219
other	6,293
referred to	4,384
followed	2,256

Table 1: Multi-Class Target Distribution

Total Documents	25,152
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3.2. Natural Language Processing (NLP) (Phase 2)

A robust NLP function (`text_preprocessing_full`) was implemented to prepare the text for vectorization.

- **Cleaning Steps:**
 1. **Lowercasing and Punctuation/Number Removal** (using `re`).
 2. **Stop Word Removal** (using `nlTK.corpus.stopwords`) to filter out high-frequency, low-value words (e.g., "the," "is," "and").
 3. **Lemmatization** (using `nlTK.stem.WordNetLemmatizer`), the most crucial step, to reduce various word forms to their semantic root.

3.3. Feature Engineering (Phase 3)

The cleaned text was converted into a numerical matrix.

- **Vectorization:** A **TF-IDF Vectorizer** was used with a parameter limit of `max_features=5000`. This creates a weighted numerical representation where the importance of a word increases if it is rare across the corpus but frequent within a single document.
- **Output Shape:** The resulting feature matrix was **(25152, 5000)**.
- **Data Splitting:** The data was partitioned into an 80% training set (N=20,121) and a 20% testing set (N=5,031), employing the **stratify** parameter to ensure the class proportions were maintained in both subsets.

3.4. Model Training and Selection (Phase 4)

- **Model: Logistic Regression** (`sklearn.linear_model`) was chosen for its interpretability and efficacy as a linear multi-class classifier.
 - **Training:** The model was fitted using the training features and the multi-class labels.
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4. Results

The model's performance was evaluated using standard multi-class metrics on the test set.

Overall Performance

Metric	Score
Overall Accuracy	0.55
Weighted Avg. F1-Score	0.50

Multi-Class Classification Report

This report provides detailed metrics for each of the four classes:

Class	Precision	Recall	F1-Score	Support (Test Set)
cited	0.57	0.85	0.68	2,444
followed	0.76	0.08	0.15	451
other	0.46	0.39	0.42	1,259
referred to	0.62	0.21	0.31	877

Confusion Matrix

The Confusion Matrix (Table 2) visually represents the model's predictive performance, showing where the true labels fall against the predicted labels.

Table 2: Confusion Matrix				
Actual	Predicted: cited	Predicted: other	Predicted: followed	Predicted: referred to
Actual: cited	2071	313	4	56
Actual: other	720	486	7	46
Actual: followed	295	105	38	13
Actual: referred to	541	150	1	185

5. Analysis and Discussion

A. Performance Strengths

The model demonstrated a strong capability in handling the most frequent class, **cited**.

- **High Recall for cited (0.85):** The model correctly identified 85% of all documents that were truly of the cited type. This success validates the effectiveness of the **Enhanced NLP** (Lemmatization and Stop Word Removal) in isolating the common linguistic patterns associated with general citation.

B. Critical Limitation: Data Imbalance and Bias

The model's overall performance is critically limited by the data imbalance problem, leading to a strong **majority class bias**.

- **Bias Confirmation:** The model learned that predicting the cited class is the statistically safest choice. This is evident in the Confusion Matrix:
 - **False Negatives (FN) for followed:** Out of 451 actual followed cases, the model incorrectly classified **295** as the majority class, cited.
 - **Low Recall for Minority Classes:** This bias crushed the ability to correctly identify the crucial legal relationships: **followed (Recall 0.08)** and **referred to (Recall 0.21)**. The model is essentially unable to distinguish these subtle differences.
- **Impact on Legal Application:** A Legal Research Assistant with an 8% recall rate for "followed" cases is unusable, as it would fail to flag the most important judicial adoptions of precedent.

C. Addressing the Imbalance

The project successfully identified the challenge, setting the stage for advanced solutions. The current limitation is not due to a poor methodology, but rather a direct result of the nature of the data and the model type selected for a baseline.

6. Conclusions

This project successfully implemented a full **Multi-Class Legal Citation Classifier**, fulfilling all technical requirements for the Python for AI and DS course. The solution demonstrated mastery over core fundamentals: **Data Preparation, Enhanced NLP, TF-IDF Feature Engineering**, and **Multi-Class Logistic Regression**.

The model achieved an overall accuracy of **55%**. While successful in building the pipeline, the project's key finding is a practical constraint: basic machine learning models struggle severely when deployed on highly imbalanced, nuanced legal text. The final model functions as a **validated Level 1 component** of a larger system.

7. References (Placeholder)

- Kaggle Dataset: Law Field Classification dataset.
- Python Libraries: Pandas, NumPy, NLTK, Scikit-learn, Matplotlib, Seaborn.